

Biostatistics

MATH 132B Semester Fall 2026

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Instructor Christopher
Saginaw Valley State University

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Course Documents

Syllabus

Instructor Information

Instructor	Christopher Servant
Pronouns	he/him
Phone Number	(989) 964-4894
Email	cservant@svsu.edu
Office	SW 340
Student Hours	

Monday	1:00 PM - 2:00 PM
Tuesday	12:00 PM - 2:00 PM
Wednesday	1:00 PM - 2:00 PM

Note: Please do not contact your instructor via Canvas. Only use his email or phone above.

Note: If you need a more private appointment to discuss grades or another personal matter, additional student hours can be made to meet in my office or via ZOOM.

Course Information

Course Dates	August 31, 2026 - December 19, 2026
Meeting Times	M, W: 10:30 AM - 12:20 PM
Location	Brown Hall 115
Credits	4
Prerequisites	All Math Basic Skills or equivalent must be completed prior to taking this course. This is done by passing MATH 082 or MATH 072 with a C or higher.

Course Description

In this introduction to statistical methodology, students identify methods of acquiring, organizing and analyzing data so as to make decisions and respond to arguments. Topics include graphical representation of data, probability

distributions, confidence intervals and hypothesis test, parametric and non-parametric techniques, regression and correlation, and analysis of variance. Special emphasis on applications from biology and the health sciences.

Course Materials

Text	<i>Introductory Statistics</i> by Hannah Seidler-Wright You can view an online copy of this free, open source textbook at Introductory Statistics .
Software	A statistical program called R via R studio to perform statistical calculations in this course. This is free software. There will tutorial videos and lectures provided to learn this program.
Three Ring Binder	This will be helpful to keep your notes and class worksheets in.

Online Course Component

In this class, you will have access to course notes, lecture videos, handouts, deadlines, and announcements on Canvas. If you are absent or class is canceled due to weather or other circumstances, it is your responsibility to log on and obtain missed notes, work, etc. Grades will be updated periodically on Canvas. It is also important to check your SVSU email regularly.

Course Outcomes

This General Education course introduces fundamental algebraic terminology and skills; demonstrates how ingenuity and creativity are essential in developing mathematical models to solve real world problems; cultivates critical thinking and logical reasoning for effective problem solving and rational decision-making in a complex technological society; and develops communication skills for translating real world situations into symbolic mathematical relationships and for interpreting mathematical results within their real world contexts.

The following lists the set of outcomes covered in this course:

1. Basic Probabilities
2. Describing Categorical Data
3. Binomial Distribution
4. Normal and χ^2 Distributions
5. Sampling Distribution for $\hat{p}$
6. Inference for One Proportion
7. Inference for Two Proportions
8. Inference for More than Two Proportions
9. Describing Numerical Data
10. Student t and F Distributions
11. Sampling Distribution for $\bar{x}$

12. Inference for One Mean
13. Inference for Two Means
14. Inference for More than Two Means
15. Describing Two Variables
16. Correlation and Linear Regression
17. Inference for Correlation Coefficient
18. Nonparametric Tests

Evaluation

Video Notes For each section covered in class, students are required to submit completed notes based on the assigned lecture videos. One submission is required per section and is due before the start of class on the day that section is covered. Notes are graded on a 0–2 scale:

- 2:** Notes are submitted on time, are complete, and reflect engagement with the assigned videos.
- 1:** Notes are submitted on time but are missing significant portions.
- 0:** Notes are not submitted on time.

Notes are graded for completion only, not for correctness. No late submissions are accepted. Notes may be handwritten, typed, or completed on a tablet, but all submissions must be uploaded as a single PDF file. Students are expected to know how to scan or create a PDF of their notes; guidance will be provided as needed.

Worksheets Worksheets are completed during class and are intended to help students practice and apply the concepts being studied. Each worksheet is paired with a MyOpenMath assignment. Students are expected to answer the questions in MyOpenMath and show supporting work on the worksheet packet. Both the completed worksheet and completion of the associated MyOpenMath assignment are required for credit.

Students may collaborate with classmates while working on worksheets and may use their notes and classroom laptops. The use of cell phones, generative AI tools, or other unauthorized resources is not permitted. Worksheets are due at the end of class and may be submitted up until the beginning of the next class period without penalty. Worksheets submitted after the start of the next class period will not be accepted.

While correctness is primarily assessed through MyOpenMath, students are expected to show clear and meaningful work on their worksheet packets. Work that is incomplete, inconsistent with the submitted answers, or does not demonstrate an attempt to solve the problem may result in reduced credit.

Quizzes Quizzes are administered approximately every fourth class meeting throughout most of the semester. Quizzes assess a student's ability to independently apply the concepts and skills developed through class activities, worksheets, and assigned course materials.

Each quiz is scored on a 0-3 scale:

- 3:** Demonstrates a strong understanding of the outcome with little to no error. Work is complete, mathematically correct, and clearly communicates the solution process.
- 2:** Demonstrates substantial understanding of the outcome but contains one or more minor errors. The overall solution process is appropriate and most of the work is correct.
- 1:** Demonstrates partial understanding of the outcome but contains significant errors or omissions. Some correct work is shown, but the student has not yet demonstrated competency.
- 0:** The outcome is not attempted or the work submitted does not demonstrate an understanding of the outcome.

Each quiz includes two in-class attempts. For each quiz, the highest score earned across all attempts is used when calculating the Quiz portion of the course grade.

A third attempt is available only after attending tutoring. Students must work with a tutor to review the relevant worksheet problems and their first two attempts before a third attempt will be granted.

Quizzes are not given as make-ups. The multiple-attempt structure is intended to accommodate absences and scheduling conflicts. Students who miss an in-class attempt may use a remaining attempt to demonstrate competency.

For the final quizzes of the course, only one in-class attempt may be provided during the final exam period. Additional attempts must be completed outside of class time by appointment.

Lab Reports Students will complete three lab reports during the semester. Each report requires students to locate an appropriate data set, analyze the data using the statistical methods studied in the course, and communicate their findings clearly. The source of the data set must be identified in the report.

The three lab reports have the following data requirements:

1. *Inference for Numerical Data.*

The data set must contain a quantitative response variable and a categorical explanatory variable with at least three groups. There must be enough observations in each group to perform an ANOVA and evaluate the conditions required for the analysis.

2. *Inference for Categorical Data.*

The data set must contain at least two categorical variables. Each variable must have at least two categories so that the data can be organized into a contingency table of at least size 2×2 . There must be enough observations to perform an appropriate chi-square test.

3. *Regression.*

The data set must contain at least two quantitative variables with paired observations. There must be enough observations to construct and analyze a regression model.

Each lab report must contain the following components:

1. *Data Set Summary.*

In no more than one page, describe the source and context of the data, identify the variables being studied, explain how the data were collected when that information is available, and state the question or questions being investigated.

2. *Graphical Analysis.*

Construct appropriate statistical graphics that help describe the data and address the questions being investigated. All graphics must include appropriate titles, labels, units, and legends when needed.

3. *Statistical Analysis.*

Perform the required statistical procedures and several appropriate statistical tests on the data. Include the relevant calculations, verify the conditions or assumptions for each procedure, and report the results using appropriate statistical notation.

4. *Summary of Results.*

Explain the results of the statistical procedures in the context of the data. The discussion should identify the important findings and explain what the numerical results, confidence intervals, significance tests, or models indicate.

5. *Conclusion.*

In no more than one page, summarize the princi-

Grading

Your final course grade will be determined by your performance in four categories: Video Notes, Worksheets, Quizzes, and Lab Reports. A letter grade will be determined separately for each category using the requirements below. Your final course grade will be the lowest of the four category grades.

Letter Grade	Video Notes Completed	Worksheets Completed	Quiz Points	Lab Report
A	90%–100%	90%–100%	40-45	77-90
B	80% to less than 90%	80% to less than 90%	33-39	63-76
C	70% to less than 80%	70% to less than 80%	27-32	50-62
D	60% to less than 70%	60% to less than 70%	18-26	36-49
F	Less than 60%	Less than 60%	0-17	0-35

Here are some examples showing how final course grades are determined:

1. A student completes 94% of the Video Notes, completes 92% of the Worksheets, earns 56 Quiz points, and earns 140 Lab Report points. The student earns an A in all four categories. Therefore, the student's final course grade is an A.
2. A student completes 95% of the Video Notes, completes 84% of the Worksheets, earns 55 Quiz points, and earns 135 Lab Report points. The student earns an A for Video Notes, a B for Worksheets, an A for Quizzes, and an A for Lab Reports. Since the lowest category grade is a B, the student's final course grade is a B.
3. A student completes 83% of the Video Notes, completes 84% of the Worksheets, earns 51 Quiz points, and earns 100 Lab Report points. The student earns a B for Video Notes, a B for Worksheets, a B for Quizzes, and a C for Lab Reports. Since the lowest category grade is a C, the student's final course grade is a C.

Plus and minus grades may be assigned based on a student's overall participation and effort throughout the semester, including class participation and attendance at office hours or tutoring sessions. For example, a student whose category grade is a B but who demonstrates limited participation and effort may receive a B-, while a student whose category grade is a C and who demonstrates a high level of participation and effort may receive a C+. A student who demonstrates the expected level of participation and effort will receive the unmodified letter grade. SVSU does not award C-, D+, or D- grades, so these grades will not be assigned.

Tips for Success

1. Read the textbook before attending class. The instructors explanations will make more sense and when they don't you will be more likely able to ask a relevant question.
2. Attend each class session, both physically and mentally.
3. Read the syllabus and look over the schedule on the last page of the syllabus.

4. Build community with other students in the course and create study groups.
5. Work outside of class. Math is learned by doing not seeing.
6. Ask for help. Stop in to see me, email me, call me, or ask others in class for help.
7. For free math tutoring; find information about the Math & Physics Learning Center at [MPLC](#)

Academic Integrity Policy

In MATH 132B, any occurrence of cheating or academic dishonesty may be officially documented. The student or students involved may then receive a copy of the documentation. The first occurrence may result in failure of the coursework and will be reported to the Office of Student Conduct Programs for the record. Any additional occurrence will be reported to the Office of Student Conduct Programs for adjudication.

You may work in groups or with tutors on homework and lab reports when permitted, but you must complete your own work. Quizzes must be completed independently without resources. Inappropriate use of laptops, cell phones, calculators, tablets, websites, outside assistance, or other resources is considered cheating. If you are unsure whether something is permitted, ask your instructor before using it.

See [SVSU's Academic Integrity Policy](#) for additional information.

Withdrawal Policy

Please consult an academic advisor and the Student Financial Services Center before withdrawing from any course. They can help you understand the possible effects that withdrawing may have on your degree progress and financial aid.

See [SVSU's withdrawal information](#) for additional details.

Important Dates

- September 4, 2026, is the last day to withdraw with a 100% refund and have the course deleted.
- September 18, 2026, is the last day to withdraw with a 100% refund and a "W" grade.
- October 2, 2026, is the last day to withdraw with a 50% refund and a "W" grade.
- November 13, 2026, is the last day to withdraw with a "W" grade.

Disability Statement

Students with disabilities that may restrict their full participation in course activities are encouraged to meet with the instructor or contact the SVSU Office of Accessibility Resources and Accommodations, located in Wickes 260, at 964-7000 for assistance.

Non-Discrimination Statement

SVSU does not discriminate based on race, religion, color, gender, sexual orientation, national origin, age, physical impairment, disability, or veteran status in the provision of education, employment, and other services.

Land Acknowledgment Statement

At Saginaw Valley State University, we recognize that we are located on the ancestral homelands of the Anishinaabe, Wyandot, and Sauk. We specifically acknowledge the Saginaw Chippewa Tribe and the land ceded in the Saginaw Treaty of 1819, on which we reside. We appreciate the importance of preserving and honoring the traditions of Indigenous People who are an integral part of our shared history. As we pledge to serve as responsible stewards of the land, we endeavor for this history to inform our teaching, scholarship, and commitment to community.

Tentative Syllabus

This syllabus and course schedule are subject to change if class needs warrant changes due to illness, inclement weather, or other circumstances.

Handouts

Introduction to Probability

Outcomes

- 1) Distinguish between empirical and theoretical probability.
- 2) Explain the law of large numbers.
- 3) Identify sample spaces and events.
- 4) Calculate and interpret basic probabilities.

Definitons

Definition 1 Sample Space.



Definition 2 Events.



Example 3 Seed Germination. A biologist plants three seeds labeled A , B , and C . After seven days, each seed has either germinated or not germinated. An outcome records the results for seeds A , B , and C , in that order.

1. Identify the sample space.
2. Identify the event that exactly two seeds germinate.



Definition 4 Empirical Probability.



Definition 5 Theoretical Probability.



Example 6 Pea Plant Genetics. In a simplified model of pea plant genetics, each parent has one dominant allele P and one recessive allele p . An offspring receives one allele from each parent. The four equally likely allele combinations are PP , Pp , pP , and pp . A plant has purple flowers if it receives at least one P .

A scientist crosses the two plants and grows 80 offspring. Of these, 57 have purple flowers.

1. Find the theoretical probability that an offspring has purple flowers.
2. Find the empirical probability that an offspring has purple flowerslaw-large-num.



Question 7 Why do the two answers above not match? ▲

Definition 8 Law of Large Numbers. ♦

Exmaples

Example 9

		Die B					
		1	2	3	4	5	6
Die A	1						
	2						
	3						
	4						
	5						
	6						

Consider rolling two fair six-sided dice and recording the sum of the numbers rolled. Find the probability of obtaining each sum from 1 through 12. Write each probability first as a reduced fraction and then as a decimal rounded to four decimal places. ▲